SE444 Makeup Mid-Term II – Answer Key & Grading Guide (Fall 2025)

Part I: Multiple Choice (4 pts)

Option	Q1 (1pt)	Q2 (1pt)	Q3 (1pt)	Q4 (1pt)
A	**X**		**X**	
B			**X**	**X**
C	**X**	**X**	**X**	
D				**X**
E			**X**	

Answers: Q1: A,C — Q2: C — Q3: A,B,C,E — Q4: B,D **Grading:** 1pt if all correct; 0.5pt if 1 error; 0pt if 2+ errors.

Part II: Logical Reasoning (10 pts)

Q II.1: FOL Translation (3 pts)

Answer:

$$\forall p, d, h\,(\text{Patient}(p) \land \text{Doctor}(d) \land \text{Hospital}(h)$$
$$\land \text{TreatedBy}(p, d) \land \text{WorksAt}(d, h)) \rightarrow$$
$$(\exists m, dis\,(\text{Medication}(m) \land \text{Disease}(dis)$$
$$\land \text{Prescribes}(d, p, m) \land \text{HasDisease}(p, dis) \land \text{Treats}(m, dis)))$$

Rubric: 3pts correct; 2pts minor errors; 1pt partial; 0pts wrong structure.
Key: $\forall p, d, h$; implication; $\exists m, dis$; all predicates.

Q II.2: FOL with Negation (3 pts)

Answer (accept any equivalent):
A: $\forall d, p, m\,((\text{Doctor}(d) \land \text{Patient}(p) \land \text{Medication}(m) \land \text{Prescribes}(d, p, m)) \rightarrow \text{Examined}(d, p))$
B: $\forall d, p, m\,(\text{Doctor}(d) \land \text{Patient}(p) \land \text{Medication}(m) \land \neg\text{Examined}(d, p) \rightarrow \neg\text{Prescribes}(d, p, m))$
Rubric: 3pts any equivalent form; 2pts minor errors; 1pt correct idea; 0pts wrong.

Q II.3: CNF Conversion (4 pts)

Steps (0.5pt each, max 4pts):
1. Negate: $\neg\forall x\,(\text{Doctor}(x) \rightarrow \exists y\,(\text{Patient}(y) \land \text{TreatedBy}(y, x)))$
2. Move $\neg$ in: $\exists x\,\neg(\text{Doctor}(x) \rightarrow \exists y\,(\ldots))$
3. Eliminate $\rightarrow$: $\exists x\,(\text{Doctor}(x) \land \neg\exists y\,(\ldots))$
4. Push $\neg\exists$ in: $\exists x\,(\text{Doctor}(x) \land \forall y\,(\neg\text{Patient}(y) \lor \neg\text{TreatedBy}(y, x)))$
5. Skolemize $\exists x$ with a: $\text{Doctor}(a) \land \forall y\,(\neg\text{Patient}(y) \lor \neg\text{TreatedBy}(y, a))$
6. Drop $\forall y$: CNF $= \{\text{Doctor}(a)\}, \{\neg\text{Patient}(y), \neg\text{TreatedBy}(y, a)\}$
Rubric: 1pt negation+$\neg\forall$; 1pt implication+DeMorgan; 1pt $\neg\exists$+inner DeMorgan; 1pt Skolem+CNF.

Part III: CSP (6 pts)

Q III.1: Formulation (2 pts)

Variables (0.5pt): R_1, R_2, R_3
Domains (1pt):
- $D(R_1) = \{\text{Smith}, \text{Brown}\}$
- $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$
- $D(R_3) = \{\text{Jones}, \text{Brown}\}$

Constraints (0.5pt):
- $R_1 \neq R_2$ (overlap 10:30–11:00)
- $R_2 \neq R_3$ (overlap 11:00–11:30)

Q III.2: AC-3 Algorithm (3 pts)

Answer: NO PRUNING - All domains unchanged
AC-3 Algorithm Execution:
Initial State:
- $D(R_1) = \{\text{Smith}, \text{Brown}\}$
- $D(R_2) = \{\text{Smith}, \text{Jones}, \text{Brown}\}$
- $D(R_3) = \{\text{Jones}, \text{Brown}\}$
- Queue: (R_1, R_2), (R_2, R_1), (R_2, R_3), (R_3, R_2)

Step-by-step Arc Processing:
Arc 1: $(R_1 \to R_2)$ **- Constraint** $R_1 \neq R_2$
- Check R_1 = Smith: Needs $R_2 \neq$ Smith $\to$ R_2 = Jones, Brown ✓Supported
- Check R_1 = Brown: Needs $R_2 \neq$ Brown $\to$ R_2 = Smith, Jones ✓Supported

Result: No values pruned from $D(R_1)$
Arc 2: $(R_2 \to R_1)$ **- Constraint** $R_1 \neq R_2$
- All values in $D(R_2)$ have support in $D(R_1)$

Result: No values pruned from $D(R_2)$
Arc 3: $(R_2 \to R_3)$ **- Constraint** $R_2 \neq R_3$
- All values in $D(R_2)$ have support in $D(R_3)$

Result: No values pruned from $D(R_2)$
Arc 4: $(R_3 \to R_2)$ **- Constraint** $R_2 \neq R_3$
- All values in $D(R_3)$ have support in $D(R_2)$

Result: No values pruned from $D(R_3)$
Final State: All domains unchanged - AC-3 terminates
Grading Rubric (3 pts):
- **2.0 pts:** Correct answer (no pruning) + shows systematic arc checking (at least 3 arcs)
- **1.5 pts:** Correct answer + shows checking of 2 arcs with support analysis
- **1.0 pt:** Correct answer + minimal arc checking (1 arc) or just states final domains
- **0.5 pt:** Incorrect pruning but shows logical AC-3 process
- **0 pts:** Only final answer without any work shown

Key: Must show arc-by-arc checking process, not just final result.

Q III.3: Valid Solution (1 pt)

Accept ANY valid assignment that satisfies:
- $R_1 \in \{\text{Smith}, \text{Brown}\}$, $R_2 \in \{\text{Smith}, \text{Jones}, \text{Brown}\}$, $R_3 \in \{\text{Jones}, \text{Brown}\}$
- $R_1 \neq R_2$ (Rooms 1&2 overlap)
- $R_2 \neq R_3$ (Rooms 2&3 overlap)

All Valid Solutions:

Solution	R_1	R_2	R_3
1	Smith	Jones	Brown
2	Smith	Brown	Jones
3	Brown	Smith	Jones
4	Brown	Jones	Brown

Grading:
- **1 pt:** Any valid assignment
- **0 pts:** Invalid (violates constraints or domains)

Verification: Check domains + $R_1 \neq R_2$ + $R_2 \neq R_3$